1. Given an array arr[] of non-negative numbers. Each number tells you the maximum number of steps you can jump forward from that position.

Input: arr[] = [1, 3, 5, 8, 9, 2, 6, 7, 6, 8, 9]
Output: 3
Explanation: First jump from 1st element to 2nd element with value 3. From here we jump to 5th element with value 9, and from here we will jump to the last.

Input: arr = [1, 4, 3, 2, 6, 7]
Output: 2

Input: arr = [0, 10, 20]
Output: -1

2. Given an array arr[] of size n consisting of non-negative integers, where each element represents the height of a bar in an elevation map and the width of each bar is 1, determine the total amount of water that can be trapped between the bars after it rains.

Input: arr[] = [3, 0, 2, 0, 4]
Output: 7
Explanation: We trap 0 + 3 + 1 + 3 + 0 = 7 units.

Input: arr[] = [3, 0, 1, 0, 4, 0, 2]
Output: 10

Input: arr[] = [1, 2, 3, 4]
Output: 0

3. Given an array arr[], where arr[i] represents the number of pages in the i-th book, and an integer k denoting the total number of students, allocate all books to the students such that:
Each student gets at least one book.
Books are allocated in a contiguous sequence.
The maximum number of pages assigned to any student is minimized.
If it is not possible to allocate all books among k students under these conditions, return -1.

Input: arr[] = [12, 34, 67, 90], k = 2
Output: 113
Explanation: Books can be distributed in following ways:

[12] and [34, 67, 90] - The maximum pages assigned to a student is 34 + 67 + 90 = 191.
[12, 34] and [67, 90] - The maximum pages assigned to a student is 67 + 90 = 157.
[12, 34, 67] and [90] - The maximum pages assigned to a student is 12 + 34 + 67 = 113.
The third combination has the minimum pages assigned to a student which is 113.

Input: arr[] = [15, 17, 20], k = 5
Output: -1

Input: arr[] = [22, 23, 67], k = 1
Output: 112

4. Given a string of digits (’1’ to ’9’), return the number of ways to decode it (e.g., ’A’ → 1, ’B’ → 2, ..., ’Z’ →26).

Input: "226"
Output: 3 (Decode as "BZ", "VF", or "BBF").

5. Given a strictly sorted 2D matrix mat[][] of size n x m and a number x. Find whether the number x is present in the matrix or not.
Note: In a strictly sorted matrix, each row is sorted in strictly increasing order, and the first element of the ith row (i!=0) is greater than the last element of the (i-1)th row. Must complete the problem in O(log(mn)) where m and n are row and column respectively.

Input: mat[][] = [[1, 5, 9], [14, 20, 21], [30, 34, 43]], x = 14
Output: true

Input: mat[][] = [[1, 5, 9, 11], [14, 20, 21, 26], [30, 34, 43, 50]], x = 42
Output: false